Architects of the Mind: The Science of the Build

Connecting brain anatomy to technology

Name: _____

Date: _____ Grade: _____

👋 Let's Start!

In this activity, you'll learn how your brain works when you use technology. You will look at different parts of the brain and how they connect to things like gaming and creating apps. Get ready to explore the science of your brain!

- **Dopamine:** A chemical in your brain that makes you feel good. It can come from easy things or from hard work.
  - *Example: Eating a sweet treat gives you a quick burst of dopamine.*
- **Iterative Design:** Making something better step by step. You test, then change, then test again.
  - *Example: Making a game and fixing bugs until it's fun to play.*
- **Neural:** Having to do with your brain. Your brain cells are called neurons.
  - *Example: Neural pathways are like roads in your brain.*

Tips for Success

Think about your own experiences when you connect your brain parts to tech. There are no wrong answers as long as you explain your thinking.

📖 Read the Following Text

Read the following text carefully. You will use it to complete the tasks that follow.

The Amazing Brain and Technology

Your brain is like a super computer. It controls everything you do. Different parts of your brain do different jobs. When you use tech, many parts work together.

The **frontal lobe** helps you think and plan. When you play a game, it helps you make choices. The **parietal lobe** helps you feel touch and know where things are. When you use a screen, it helps you move your fingers.

The **temporal lobe** helps you remember things and understand language. It helps you recall stories from games. The **occipital lobe** helps you see. When you watch videos, it is working hard.

The **cerebellum** helps you balance and move. When you use a controller, it helps you push the right buttons. The **brain stem** keeps you alive by controlling basics like breathing. Even when you are focused on a game, your brain stem is working.

Doing hard things, like coding or making art, can give your brain “earned joy”. This is better than “cheap dopamine” from things like scrolling. Earned joy makes your brain stronger.

🔗 Building Connections

Now that you've read the text, complete the table below to connect the key ideas and show why they matter.

Key Concept	Your Understanding	Why is this important?
Frontal Lobe	*Thinking and planning*	*Helps you make good choices when using tech*
Parietal Lobe		
Temporal Lobe		
Occipital Lobe		
Cerebellum		
Brain Stem		

✍️ Try It Out

Use what you learned to answer the question below.

The Scroll (Cheap Dopamine)	The Build (Earned Joy)
Watching auto-play videos	Solving a code error

1. Which activity makes your brain feel stronger and why?

2. **For each brain part, write a Tech Connection (How that part is used when gaming or creating):**

View the table on the next page.

Frontal Lobe: (Thinking and Planning)	
Parietal Lobe: (Touch and Space)	
Temporal Lobe: (Memory and Language)	
Occipital Lobe: (Vision)	
Cerebellum: (Balance and Movement)	
Brain Stem: (Survival Basics)	

3. **Write your own simple definition for each word:**

View the table on the next page.

Prompt Engineering	
Iterative Design	
Cheap Dopamine	
Earned Joy	

4. Circle ONE project concept. Sketch a small Icon for it in the box.

1. Wellness App: An app that helps the Frontal Lobe stop the “Infinite Scroll.”
2. AI Adventure: A game that uses Temporal Lobe memory to solve mysteries.
3. Hardware Hero: A custom controller concept that uses the Cerebellum for movement.

Sketch the Icon on the next page.

🎬 Show What You Learned

Now that you've explored the key concepts and how they connect, create a project to show your understanding in a new way.

Create a short video. Explain how the brain works when people use technology.

Follow the steps below to complete your creative output: Use Magic Media or Elements to help storyboard or record your idea.

- Plan your video: What parts of the brain will you talk about?
- Write a script: What will you say about each part of the brain?
- Record your video: Be clear and interesting.
- Share your video with the class.

Reflect on Your Knowledge Transfer

Now that you've explored how ideas can move from understanding to real-world use, take a moment to look back on how you connected concepts, recognized patterns, and applied them to new situations.

What was the most interesting thing you learned about how your brain works with technology?

How might you use what you learned to make better choices about your tech use?

What other questions do you have about the brain and technology?